

**SWINBURNE VIETNAM
ALLIANCE PROGRAM**



E-commerce Platforms for Traditional Handicrafts from Southern Vietnam

by Truong Ngoc Gia Hieu | Pham Nguyen Minh Hoang | Dinh Hoang Phuc

Department: Swinburne University of Technology

Course Name: COS10025 – Technology in an Indigenous Context Project (HCMC)

Instructor Name: Hang Sam Nang

Submission Date: 1 Jun by 23:59

Word count: 2606

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Part A

1.1 Project Overview

The Request for Proposal (RFP) initiates the project named “E-Commerce Platforms for Traditional Handicrafts from Southern Vietnam” for several key reasons: Firstly, the organization recognizes the significant role of national identity products from centuries-old traditional villages in sustainable livelihoods, allowing local people to preserve their historical cultural distinctiveness and tourists to immerse themselves in Vietnamese culture. Moreover, this project aims to promote economic development and leverage the power of e-commerce platforms to expand market access for artisan communities, allowing hardworking, skillful artisans to have higher incomes and expand their trading networks. Therefore, the Request for Proposal (RFP) implements the project within the course “Technology in an Indigenous Context Project”.

First and foremost, the project desires to generate and publish sustainable, user-friendly, and community-controlled e-commerce platforms that offer equitable market access for traditional handicrafts from Southern Vietnam. This project not only directly benefits artisan communities and preserves cultural heritage but also champions fair trade practices. According to the project, the platforms must integrate secure payment and robust logistics solutions, comprehensive capacity building and training initiatives for artisans, and culturally sensitive marketing strategies. As a result, team GAMBLE, with the guidance of tutor Hang Sam Nang, is establishing the project across Southern Vietnam, aiming to generate and publish a replicable e-commerce platform for cultural preservation and economic empowerment across the region.

1.2 Introduction to the Problem

According to the project, Southern Vietnam comprises a diverse range of artisan communities, including Indigenous and ethnic minority groups. Moreover, these people face various significant challenges in accessing broader markets for their products, which subsequently impacts livelihoods and opportunities to improve quality of life in rural areas. Consequently, the RFP organization offers this dynamic project to provide students with opportunities to enhance important soft skills, including critical thinking, complex problem-solving, teamwork, and communication within groups. Furthermore, students who participate in the “E-Commerce Platforms for Traditional Handicrafts across Southern Vietnam” project gain valuable experiences that are extremely crucial in academic performance and comprehensive development. As a result, team GAMBLE frequently collaborates with each other, allowing teammates to have a deep understanding of challenges and difficulties in rural areas, enhancing communication and complex problem-solving techniques to address the problems easily and smoothly. Subsequently, my team is initiating the project early and is on the correct track.

Part B

2.1 Phase 1: Community Consultation, Needs Assessment, and Platform Strategy

2.1.1 Description

Based on the project specifications of the first phase, the following phase requires community consultation, needs assessment, and platform strategy. E-commerce platforms must have basic functionalities, including support mechanisms required for effective and productive platforms, and security features. Additionally, platforms should suggest suitable strategies for artisan communities, supporting comprehensive capacity building programs, culturally sensitive marketing approaches, and long-term sustainability models to ensure equitable market access and the preservation of cultural heritage. Notably, the platform of the artisan communities has one outstanding capability: local people and businesses can exchange unique products and expand their network. Furthermore, generating e-commerce platforms for artisan communities across Southern Vietnam is a significant opportunity, offering a unique chance to attract international tourism, foster substantial economic growth, and significantly enhance the overall quality of life for these communities.

2.1.2 Research and Analyze

Before designing and publishing the platforms, all members of the team need to decide which province in Southern Vietnam has various artisan communities and diverse products. First and foremost, team members believe that the research and analysis phase is a critical step in any project because it serves as the foundation for making the final and right decision. During this important phase, team GAMBLE identified several provinces, each with its own unique artifacts. For example, An Giang province is recognized for Khmer brocade and palm sugar, while Ben Tre province is known for a variety of coconut-based goods. Secondly, Đồng Tháp Province excels in traditional mat weaving and distinctive Lai Vung products. Thirdly, Kiên Giang Province offers traditional rice paper and water hyacinth handicrafts, showcasing the cultural diversity and regional identity. Lastly, Vĩnh Long Province stands out for its unique red pottery. As a result, team members have a comprehensive understanding of the distinct artisan offerings across these Southern Vietnamese provinces, allowing them to make an informed decision on which province to focus their platform design and publishing efforts.

2.1.3 Decision

After collecting crucial information, team members noticed two outstanding provinces among the list: Kien Giang and Dong Thap Provinces. However, after discussing together and considering key factors such as tourist attractions, culturally resonant historical narratives, economy, and digital literacy levels, my team made the final decision to select Kien Giang Province due to its superior tourist attraction and robust digital literacy levels, which we believe will be more conducive to our project's success.

2.2 Phase 2: Platform Design and Development

2.2.1 Description

According to the requirements of the second phase in the detailed document, my team has to design and implement not only a user-friendly interface but also core, basic functionalities, including product listing, order management, robust security, local and global language functionality implementation, logistics and shipping integration, and mobile responsiveness. Due to these second-phase requirements, my team decided to implement three interfaces: an administrative interface, a combined interface for users, consumers, and buyers, and a dedicated interface for artisan communities, via both a dedicated mobile application (for Android and iOS) and a responsive web application. Furthermore, these platforms are essential for maximizing reach and usability within the diverse artisan communities of Southern Vietnam, accommodating varying levels of digital literacy and device availability. For this reason, integrating useful workshops and training for local people is crucial to ensure effective adoption and utilization of the platforms, reducing the digital literacy gap, empowering artisan communities' quality of life, and boosting the national economy in general and the province in particular.

2.2.2 Digital Literacy Level

Based on the research and analysis phase, the results show that the digital literacy landscape, much like other provinces across Southern Vietnam, presents a mixed picture. While urban areas and younger generations swiftly adopt remarkable and cutting-edge technologies, rural areas and older demographics often face challenges. These significant challenges limit people's access to high-speed internet, a lack of affordable devices, and, crucially, a varying degree of foundational digital skills. Subsequently, dramatic efforts are being launched to improve digital literacy across the province through government initiatives, community programs and workshops, and educational outreach, aiming to empower all residents to participate fully in the digital economy and societal transformation.

For example, in Kien Giang Province, the digital transformation of education is notably progressing through the implementation of the KAV Open School model. This initiative, spearheaded by the Department of Education and Training, offers a fundamental change to traditional learning methods by leveraging information technologies. Despite a wide range of challenges in deploying IT infrastructure to remote island schools, significant government commitment has ensured internet access and ongoing facility investments. Currently, the KAV Open School enables flexible online study for students and efficient assignment/evaluation for teachers, ultimately striving to improve overall learning quality across the province.

Additionally, Kien Giang province in Vietnam is actively undergoing digital transformation as part of Project 06, demonstrating significant progress in various sectors. As of September 9, 2024, the province has observed an increase in six out of thirteen digital transformation indicators. Key achievements include a 64.2% rate of online administrative records, with 72.77% of administrative procedure records and results digitized, and a 99% synchronization rate of records on the National Public Service Portal. Additionally, healthcare facilities are integrating data for certificates and notices with the Health Insurance Appraisal Information System, and 97.4% of fishing vessels have updated their information on the national fisheries database

(VNFishbase).

Furthermore, Kien Giang province is actively boosting digital literacy, seen in the high rates of online administrative procedures and data synchronization on national platforms. Beyond just skills, the launch of a dedicated Cybersecurity and Information Operation Centre plays a vital role in creating a secure digital environment. This foundation of safety and trust is essential for encouraging wider public engagement and ultimately advancing the province's overall digital literacy levels.

2.2.3 Core Functionalities

According to the requirements of the first and second phases in the document, my team is establishing comprehensive e-commerce ecosystems for both mobile and web applications. For further details, the platforms facilitate user-friendly interfaces, product listing and browsing, robust order management from creation to delivery, filtering and sorting based on various factors, logistics and shopping solutions, multi-language support, and secure payment processing. As a result, the above core functionalities ensure that administrative, customer/buyer, and artisan communities' interfaces run smoothly and securely, fostering trust and efficiency across all user interactions.

2.2.4 Platform Technical Specifications and Storing Data

Based on the project requirements, for web application, my team utilizes modern web technologies such as HTML5, CSS3, and JavaScript for the front-end development, augmented by robust frameworks like React, Angular, or Vue.js for creating dynamic and user-friendly interfaces. Furthermore, on the backend, programming languages such as Python (with Django/Flask), Node.js (with Express), or PHP (with Laravel) will be used for server-side logic, API development, and database management, ensuring scalability and security. In the other hand, for the mobile applications, native development will employ Kotlin or Java for Android, and Swift or Objective-C for iOS, providing optimal performance and user experience. Alternatively, cross-platform frameworks like React Native or Flutter could be considered for efficient development across both mobile operating systems.

For storing data, the e-commerce platforms will leverage MySQL as its relational database management system. From my team perspective, MySQL is a suitable language to store data due to its robustness, scalability, and widespread support, making it ideal for managing large volumes of structured data. Consequently, proficiency language persists critical information such as product details, user profiles, order history, transaction records, and artisan community data, ensuring efficient retrieval and integrity across all applications.

2.2.5 Administrative Platform

The Administrative interface serves as the central control panel for the organization and designated community leaders, allowing administrators to manage, ensure oversight, and long-term sustainability of the e-commerce platform ecosystem:

- Platform Monitoring and Maintenance:

- Ability to monitor overall platform security, stability, and performance.
- Tools for ongoing maintenance and updates to ensure optimal operation.
- User Management:
 - Management of artisan accounts (approval, suspension, support).
 - Oversight of customer accounts (if applicable for support or dispute resolution).
- Content and Product Oversight:
 - Review and approval mechanisms for product listings and descriptions to ensure cultural authenticity and ethical representation.
 - Ability to manage categories, featured products, and promotional content.
- Financial Monitoring and Reporting:
 - Access to sales data, transaction logs, and payout information for transparency.
 - Generation of reports on platform performance, revenue, and artisan earnings.
- Logistics and Payment Integration Management:
 - Configuration and monitoring of integrated secure payment gateways.
 - Management of shipping provider integrations and logistics parameters.
- Community Management & Guidelines:
 - Tools to develop, implement, and enforce community management guidelines and ethical protocols.
 - Support for knowledge transfer initiatives for community members managing the platform.
- Sustainability Planning Tools: Features to track and analyze data relevant to the long-term financial and operational sustainability plan (e.g., commission models, community contributions).

2.2.6 Customer/Buyer Platform

Based on the requirements of the project in the first and second phase, my group design this platform for customer/buyer specifically, providing user-friendly interface, list products, secure,

and culturally immersive shopping experience for national and international customers in traditional Vietnamese handicrafts.

- Intuitive User Interface (UI):
 - Clean, accessible, and responsive design optimized for various devices (desktop, tablet, mobile).
 - Easy navigation and clear product categorization.
- Product Discovery and Browsing:
 - Search functionality with filters (e.g., craft type, region, price range).
 - Detailed product pages with high-quality images, compelling descriptions, and artisan stories.
 - Display of cultural origins and significance of handicrafts.
- Secure Shopping Cart and Checkout:
 - Ability to add/remove items from a shopping cart.
 - Streamlined, secure checkout process with multiple payment options commonly used in Vietnam and accessible internationally.
- Multi-Language Support:
 - Primary support for Vietnamese.
 - Secondary support for English and potentially other relevant languages to broaden reach.
- Order Tracking and History: Customers can view their order status and access past purchase history.
- Customer Support Mechanisms: Access to FAQs, contact forms, and potentially live chat or phone support for inquiries.
- Fair Trade and Ethical Transparency:
 - Clear indication of fair-trade practices and direct benefits to artisans.
 - Information about sustainable practices where applicable.

2.2.7 Artisan Communities Platform

This unique platform empowers craftsmen/craftswomen to manage their presence, exchange products, and update latest news, fostering community ownership and enhancing digital literacy. Moreover, this interesting platform serves as digital workshop and marketplace interface, enabling hard-working people directly manage unique products, sales, and interactions with customers.

- **Artisan Dashboard:** A personalized, intuitive dashboard providing a comprehensive overview of an artisan's online store performance. This includes at-a-glance metrics for sales, pending orders, product views, and overall earnings. It also serves as a central hub for accessing training materials and support resources.
- **Product Listing and Management:** An easy-to-use interface for creating and editing product listings. Artisans can upload high-quality product photography, write compelling product descriptions that convey the story and cultural significance of their crafts, and specify details such as materials, dimensions, and pricing. Robust inventory management features allow artisans to accurately track stock levels and prevent overselling.
- **Order Processing and Fulfillment:** A clear notification system alerts artisans to new orders. They can view detailed order information, update order statuses (e.g., "pending," "packed," "shipped"), and, if integrated, generate shipping labels directly from the platform. Communication tools facilitate direct interaction with customers regarding order inquiries or updates.
- **Profile Management:** Artisans can create and maintain their public profiles, which include their personal story, background in their craft, location, and any unique aspects of their traditional techniques. This feature helps customers connect with the artisans and appreciate the human element behind the handicrafts.
- **Payout and Financial Transparency:** The platform provides transparent access to financial information, including detailed breakdowns of sales, applicable commissions (if any), and direct payout schedules. This ensures artisans have clear visibility into their earnings and the financial model.
- **Training and Support Resources:** A dedicated section offers culturally appropriate and easy-to-understand training materials, such as guides, video tutorials, and FAQs. These resources cover essential topics like effective product photography, crafting engaging product descriptions, platform navigation, and best practices for customer service. Direct support channels are available for technical assistance and guidance.
- **Community Engagement Features:**
 - Beyond individual selling, this platform incorporates features designed to foster a vibrant artisan community. This may include internal communication tools, forums, or dedicated groups where artisans can share knowledge, discuss challenges, and collaborate.
 - **Inter-Artisan Product Trading:** A unique functionality allowing craftsmen and craftswomen to directly trade or exchange products, raw materials, or even specialized

tools with each other. This feature aims to facilitate resource sharing, foster collaborative creation, and strengthen the internal economy and solidarity within the artisan communities.

2.2.8 Conclusion

In conclusion, engaging in this exciting project named “E-Commerce Platforms for Traditional Handicrafts across Southern Vietnam in the course “Technology in an Indigenous Context Project – COS10025” allows team members enhance soft skills including communication, complex problem-solving, teamwork, research and analyze skills. Therefore, during the project, team GAMBLE explores diverse artifacts products from skillfull craftsmen/craftswomen across Southern Vietnam. In summary, team members not only boost soft skills which are extremely useful for academic performance but also discover interesting artisan communities across Southern Vietnam throughout the exciting project.

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